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Many Models, One City: A State Memory Framework for Integrated Urban Simulation in Nashville

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Motivation

Helping cities plan, operate, and prepare

> DAVIDSON
>
> **How Nashville's ice storm spiraled into a deadly weekslong disaster**
>
> *Inside the human toll of Nashville's two-week blackout*
>
> **Molly Davis and Kelly Puente** Nashville Tennessean
> March 10, 2026, 5:01 a.m. CT

Figure 1: The *Nashville Tennessean* on Winter Storm Fern [1].

We couple traffic, EV charging, weather, building, and power grid models to study how cities respond to different scenarios, from extreme weather to new infrastructure. Nashville serves as our first case study.

Integrating models shows the full impact of a scenario

Urban systems such as traffic, buildings, and the power grid depend on each other, yet most models simulate only one of them. Coupling the models reveals how a change in one system affects the others.

Geospatial models need a new framework

Our models exchange **geospatial entities**: real-world objects with a location, a time, and properties, such as an EV. To our knowledge, no flexible framework exists that exchanges such entities between models in a shared space and time.

Objectives

ORIGIN (**O**ak **R**idge **I**ntegrated **G**rid, **I**nfrastructure, and **N**etwork Model) has two goals:

1. Develop ORIGIN as a flexible, fast, reusable, and spatial framework
2. Demonstrate it with ORIGIN: Nashville, our first case study

Methods

ORIGIN: Flexible, fast, reusable, and spatial integration

ORIGIN supports two kinds of models: **live models**, which it runs step by step during the simulation, and **pre-run models**, which run beforehand and whose results are loaded into the **State Memory**, ORIGIN's shared database.

Methods (continued)

Flexible

The requirements on live models are minimal: the **Supervisory Model**, which coordinates all models, must be able to **write their inputs**, **advance them by one model timestep**, and **read their outputs**. Models can run on one computer or spread across several.

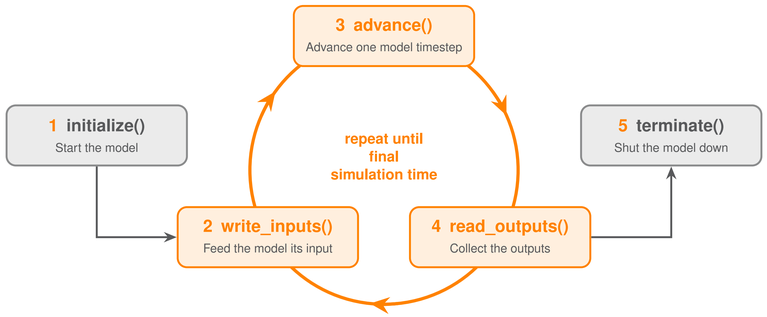


Figure 2: Life cycle of a model in the ORIGIN framework. The Supervisory Model repeatedly calls steps 2–4 every model timestep until the simulation ends.

Fast setup

A live model defines only four components: one or more **Outputs** and **Inputs**, one **Adapter**, and one **ModelSpec**. A pre-run model only defines its Outputs.

Reusable

The **Supervisory Model**, **Adapter Worker**, **Message Broker**, and **State Memory** are model-agnostic: they work with any model without changes.

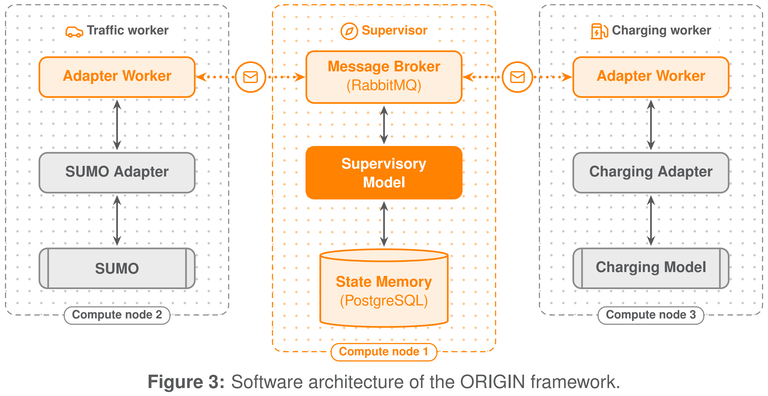


Figure 3: Software architecture of the ORIGIN framework.

Spatial *(in development)*

The State Memory places all entities in a shared space and time and builds each model's inputs from it.

Methods (continued)

ORIGIN: Nashville couples five urban models

The models currently cover a test area in Nashville and will later cover all of Davidson County, TN:

- **Traffic:** SUMO (Simulation of Urban Mobility)
- **EV charging:** ORNL-developed model
- **Building:** AutoBEM (Automatic Building Energy Modeling)
- **Weather:** WRF (Weather Research and Forecasting)
- **Power grid:** ORNL-developed model

Results (preliminary)

We ran our first 24-hour scenario in the Nashville test area, showing that ORIGIN can couple models and exchange data between them. It tested the impact of new businesses: a corporate office, a data center, and a distribution facility.

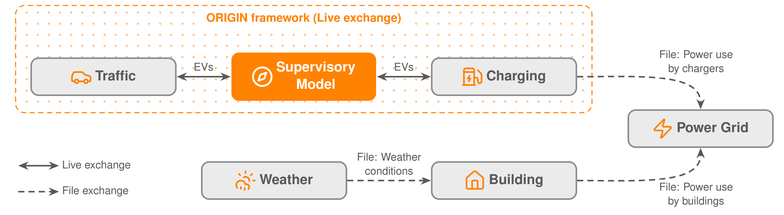


Figure 4: Model coupling in the new-business scenario.

Conclusion & Future Work

We built a first version of the flexible, fast, and reusable ORIGIN framework and successfully ran it on a Nashville test area. Next, we will:

1. Develop the framework further and finalize the State Memory's design
2. Run more scenarios, including a 3-day heat wave and Winter Storm Fern
3. Benchmark ORIGIN, possibly against other coupling approaches

Follow ORIGIN

github.com/gaberm/origin-framework

References

[1] M. Davis and K. Puente, "How Nashville's ice storm spiraled into a deadly weekslong disaster," *The Tennessean*, Mar. 10, 2026. Accessed: Oct. 14, 2026. https://www.tennessean.com/story/news/local/davidson/2026/03/10/nashville-winter-storm-power-outage-timeline/88478822007/

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